

Q1: Discuss the factors contributing to inflation in the economy. Highlight its socio-economic impacts of inflation on various segments of society. (10 marks, 150 words)

Approach to the answer:

Introduction: Start with definition of inflation.

Body: Address the demand by discussing the factors contributing to inflation followed by socio-economic impact.

Conclusion: Conclude with summarizing the topic.

Answer: Inflation refers to a continuous rise in the general price level of goods and services which affects economies through various factors, leading to significant socio-economic impacts.

Factors Contributing to Inflation:

- **Supply-Side Shocks:** Disruptions in supply chains and natural disasters cause shortages, leading to cost-push inflation. *Example: Russia-Ukraine war and monsoon failure.*
- **Imported Inflation:** Dependence on oil imports makes India vulnerable to global oil price fluctuations. *Example: High crude oil prices affecting the economy.*
- **Global Commodity Prices:** Fluctuations in key commodity prices like steel, fertilizers, etc drive inflation across sectors.
- **Currency Depreciation:** Depreciation in currency creates inflationary pressure via import-dependent sectors.
- **Monetary Policy:** Central bank's control over money supply and interest rates influences inflation through demand-supply effects. *Example: Expansionary monetary policy increasing inflation.*
- **Fiscal Policy:** Government spending, taxation, and budgetary deficits influence inflation by affecting money circulation. *Example: Austerity measures reducing inflation.*
- **Structural Factors:** Long-term issues like inadequate infrastructure and supply bottlenecks contribute to inflation. *Example: Transportation losses in agriculture commodities.*

Socio-Economic Impacts of Inflation:

1. **Social Impact:**
 - **Impact on the Poor:** Inflation disproportionately affects the poor, increasing the cost of essentials.
 - **Increase in Child Labor:** Families seek more earning members due to rising costs.
 - **Gendered Impact:** Inflation marginalizes women's interests, increasing dropouts, child marriages, and bride trafficking. *Example: Increased girl child school or college dropouts.*
 - **Poverty Trap:** Reduced investment in human capital lowers long-term social mobility. It perpetuated inter-generational poverty.
2. **Economic Impact:**
 - **Eroded Purchasing Power:** High inflation reduces consumer purchasing power, potentially leading to recession. *Example: Decreased consumer spending.*
 - **Reduced Export Competitiveness:** Inflation creates economic uncertainty, affecting business planning and investments. *Example: Stock market collapse, flight of capital.*
 - **Wealth Concentration:** Inflation impacts asset values, benefiting the rich and hurting the poor.
 - **Income Inequality:** Widened income gap due to unemployment and recession.

Addressing inflation requires coordinated policy measures and structural reforms to ensure economic stability and inclusive growth, mitigating its socio-economic impacts on society.

Q2: Discuss how the PM Formalization of Micro Food Processing Enterprises (PM FME) scheme aims to tackle the challenges encountered by micro-food processing enterprises. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with brief introduction about Micro Food processing industries.

Body: Address the demand by discussing the challenges faced by Micro-Food Enterprises then mention how PMFME scheme's role in addressing challenges.

Conclusion: Conclude with summarizing the topic.

Answer: Micro-food enterprises play a crucial role in India's food processing sector, with approximately 2.5 million such enterprises accounting for 10% of total output and 75% of jobs in the sector. Nearly 66% of these enterprises are located in rural areas.

Challenges Faced by Micro-Food Enterprises

- **Lack of Access to Credit:** Micro-food enterprises often struggle to secure funding for technology upgrades or expansion.
- **Low Investment:** These enterprises typically have low investments in plants and machinery.
- **Inadequate Infrastructure:** There is a lack of access to essential infrastructure like sorting, grading, warehousing, and cold chains.
- **Skill Shortages:** Due to limited financial resources, micro-food enterprises have difficulty attracting skilled workers.
- **Competition:** They face stiff competition from larger players who have access to superior infrastructure and resources.
- **Higher Costs:** Inability to purchase inputs in bulk leads to higher procurement costs. Additionally, they struggle to secure contracts for consistent quality inputs.
- **Poor Branding and Marketing:** These enterprises typically focus on production and often lack resources for marketing, resulting in limited market reach.

PMFME Scheme's Role in Addressing Challenges

- **Financial Support:** The scheme provides a credit-linked capital subsidy of 35% of the eligible project cost, up to a maximum of ₹10 lakh per unit for upgrading or setting up new units.
 - Additionally, seed capital of ₹40,000 per SHG member (up to ₹4 lakh per SHG) is available for working capital and purchasing small tools.
- **Infrastructure Development Support:** The scheme aids in developing common infrastructure such as incubation centers, processing facilities, warehouses, laboratories, and cold storages, benefiting enterprises like FPOs, cooperatives, SHGs, state agencies, and private enterprises.
- **Capacity Building:** Training and research support are provided in partnership with State Level Technical Institutions and private training partners to enhance the skills and knowledge of food processing enterprises, groups, and clusters.
- **Marketing and Branding:** The scheme includes provisions for marketing and branding support, with MoUs signed with NAFED and TRIFED to assist beneficiaries throughout the value chain.

The effective implementation of this scheme can provide the necessary support for micro-food enterprises, fostering their growth and development.

Q3: Although Gross domestic product (GDP) is widely utilized to assess well-being across countries, it comes with certain shortcomings. Examine. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with brief introduction about GDP.

Body: Address the demand by discussing the reasons for using GDP to compare well-being across countries then highlight the limitations of GDP as an indicator of well-being.

Conclusion: Conclude with summarizing the topic.

Answer: Gross Domestic Product (GDP) refers to the total value of all final goods and services produced within an economy over a specific period. It measures the value of production within a geographic boundary, irrespective of the nationality of the individuals or firms involved.

Reasons for Using GDP to Compare Well-Being Across Countries:

- **Economic Size and Performance:** GDP provides information about the size of an economy and its overall performance. The growth rate of real GDP is often used as an indicator of the general health of the economy.
- **Standardized Measure:** GDP is calculated consistently across countries, allowing for easy comparison and analysis of data between nations, as well as tracking changes over time.
- **Indicator of Standard of Living:** A high GDP per capita is generally associated with higher levels of consumption, income, and employment opportunities, which can enhance people's well-being.
- **State Capacity and Development:** GDP growth, as a measure of economic growth, increases the capacity of the state to fund welfare and development initiatives, such as education, healthcare, and environmental protection.

Limitations of GDP as an Indicator of Well-Being:

- **Economic Inequality:** GDP does not account for economic inequality within a country. High GDP growth can coincide with increasing income inequality, as seen in India over the past two decades.
- **Negative Externalities:** GDP does not consider the social and environmental costs of economic growth, such as pollution and resource depletion.
- **Unpaid Work Exclusion:** GDP excludes the value of unpaid work, such as childcare, eldercare, and volunteering, which contribute significantly to people's well-being.
- **Intangibles Not Measured:** GDP does not measure intangibles like leisure, quality of life, and overall happiness, which are important aspects of well-being beyond material wealth.

Due to these limitations, economists have developed alternative indices to better capture well-being, including the Human Development Index (HDI), Human Poverty Index (HPI), Gross National Happiness (GNH), and Green GDP, among others. These alternatives attempt to address the shortcomings of GDP.

Additional Information:

Alternatives to GDP

- **Human Development Index (HDI):** Measures well-being based on life expectancy, education level (mean years of schooling and expected years of schooling), and per capita income. It provides a broader perspective on development beyond economic growth.
- **Gross National Happiness (GNH):** Focuses on holistic well-being, considering factors like psychological well-being, health, education, time use, cultural diversity, and resilience.
- **Green GDP:** Adjusts GDP by accounting for environmental costs, such as pollution and resource depletion. It aims to provide a more sustainable measure of economic performance by considering ecological impacts.
- **Multidimensional Poverty Index (MPI):** Assesses poverty based on various indicators such as health, education, and standard of living.

Q4: What is digital taxation? Highlight the rationale behind implementation of digital tax in India and discuss the potential challenges associated with it. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with brief introduction about Digital taxation.

Body: Address the demand by discussing the background of digital taxation followed by rationale behind introducing digital tax in India. Also mention the challenges associated with Digital taxation.

Conclusion: Conclude with summarizing the topic or with way forward.

Answer: Digital taxation refers to the imposition of taxes on digital services and transactions. It aims to ensure that multinational technology companies pay fair taxes in countries where they operate, ultimately addressing tax avoidance and fostering equitable revenue distribution across jurisdictions.

Background:

- Digital tax refers to a levy on digital goods, services, or business activities. India was among the pioneers in this domain, introducing a 6% Equalization Levy (EL) on non-resident digital companies like Google or Facebook in 2016.
- In the Finance Act 2020, 2% Digital Service Tax (DST) levied on the trade and services of foreign e-commerce companies such as Amazon or Flipkart, provided they have an annual turnover of Rs. 2 crore or more.
- The DST ensures that non-resident digital service providers contribute taxes on revenues generated within the Indian digital market.

Rationale Behind Introducing Digital Tax in India

- **Growth of the Digital Economy:** India's digital economy, expected to reach \$200 billion by 2026, necessitates updated tax frameworks to capture revenue from digital transactions.
- **Ensuring Fair Tax Contribution:** Digital taxes ensure multinational corporations pay a fair share of taxes based on revenues from Indian users.
- **Addressing Tax Avoidance:** Digital taxation helps close loopholes exploited by large firms, preventing tax avoidance and creating a level playing field for local businesses.
- **Compliance with Global Standards:** India's digital tax aligns with global efforts, such as OECD initiatives such as Base Erosion and Profit Shifting (BEPS), to create equitable tax structures for multinational corporations.
- **Boosting Local Revenues:** Digital taxes can increase local revenues, funding public infrastructure, education, and healthcare, thereby supporting economic development and improving living standards.

Challenges Associated with Digital Tax:

- **Legal Challenges:** Digital taxes can face legal issues, particularly over international tax jurisdiction and conflicts with existing tax treaties.
- **Risk of Double Taxation:** Digital taxes may result in double taxation, as multiple countries could claim tax rights over the same income, impacting multinational corporations and deterring investment.
- **Economic Impact on Digital Firms:** Higher operational costs from digital taxes may hurt digital firms, potentially reducing investments in innovation and growth.
- **International Trade Tensions:** Unilateral digital tax policies can lead to trade tensions and retaliatory measures, potentially sparking economically harmful trade wars.
- **Increased Costs for Consumers:** There is concern that these taxes might ultimately be passed on to consumers, thereby increasing their financial burden rather than taxing the companies.

Developing clear guidelines and establishing a robust dispute resolution mechanism are essential steps. Additionally, bilateral and multilateral renegotiation of tax treaties, overriding domestic tax laws, can help resolve double taxation issues.

Q5: Highlight the importance of Micro, Small, and Medium Enterprises (MSMEs) for a healthy economy and suggest measures to support its growth.
(10 marks, 150 words)

Approach to the answer:

Introduction: Start with introducing in brief about MSME sector.

Body: Address the demand by discussing the importance of MSMEs for a healthy economy then suggest measures to support MSME growth.

Conclusion: Conclude with summarizing the topic.

Answer: Micro, Small, and Medium Enterprises (MSMEs) are crucial for a healthy economy, contributing significantly to employment, exports, and GDP while fostering inclusive and sustainable development.

Importance of MSMEs for a Healthy Economy

- **Employment Generation:** MSMEs contribute significantly to job creation, employing around 110 million people in India.
- **GDP Contribution:** MSMEs contribute approximately 30% to India's GDP, driving economic growth.
- **Innovation and Entrepreneurship:** MSMEs foster innovation and entrepreneurship, crucial for a dynamic economy.
- **Export Promotion:** MSMEs contribute about 45% of India's exports, enhancing foreign exchange earnings.
- **Inclusive Growth:** MSMEs promote regional development and reduce income disparities by operating in rural and semi-urban areas.
- **Supply Chain and Industrial Linkages:** MSMEs support large industries by providing raw materials, components, and services.

Measures to Support MSME Growth

- **Enhanced Credit Access:** Improve timely and adequate financing.
 - For example: Credit Guarantee Trust Fund, Mudra Yojna, and 59-minute loan scheme.
- **Technology Upgradation:** Provide modern technology to boost productivity and competitiveness.
 - For example: ASPIRE scheme and AI interventions in packaging firms.
- **Marketing and Branding Support:** Improve marketing and branding to scale operations.
 - For example: Establish Common Facility Centers (CFCs) for raw materials and marketing support (SFURTI scheme).
- **Regulatory Reform:** Address high compliance costs to encourage growth.
 - For example: Incentivize 'dwarf' MSMEs to expand beyond 100 workers as suggested in Economic Survey-2019.
- **Infrastructure Development:** Enhance infrastructure for smooth operations.
 - For example: Improve logistics, transportation, and connectivity for MSMEs.

MSMEs play a crucial role in economic development, fostering innovation, employment, and inclusive growth, thereby ensuring a resilient and healthy economy.

Q6: What is gender budgeting, and discuss the measures required for its effective implementation in India.
(10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Gender Budgeting.

Body: Address the demand by highlighting limitation of Gender Budgeting followed by steps for effective implementation of Gender Budgeting.

Conclusion: Conclude with summarizing the topic.

Answer: Gender budgeting is the practice of applying a gender perspective to the allocation and tracking of public funds. The primary objective is to ensure that public resources are allocated in a manner that addresses

the diverse needs of men and women. India has been releasing a Gender Budget Statement (GBS) alongside its Union Budget since 2005.

Limitations of Gender Budgeting in India:

- **Lack of Clarity in Allocation:** The current classification does not clarify the distribution of funds in schemes that allocate less than 30% of their budget to women.
- **Limited Budgetary Allocation:** India's gender budget has consistently ranged between 4-6% of total expenditure and less than 1% of its GDP.
- **Concentration in Key Ministries:** Approximately 90% of the gender budget is concentrated in five key areas such as digital literacy, skill training, and the prevention of domestic violence against women receive only 2% of the budget.
- **Insufficient Gender Data:** There is a lack of data tracking to assess who benefits from government schemes in terms of gender.

Steps for Effective Implementation of Gender Budgeting in India:

- **Gender Data Collection:** Collect and analyze gender-disaggregated data to inform budgetary decisions.
- **Capacity Building:** Training for personnel on gender sensitivity, income-generating resources, and fund allocation is essential across the country's administrations.
- **Increased Participation of Women:** Women should be actively involved in planning and budgeting processes as equal partners in decision-making and implementation, rather than merely as beneficiaries.
- **State-Level Reforms:** States should evaluate Annual Plans with a gender perspective, and form Gender Task Forces at the state, district, block, and village/Urban Local Bodies levels.
- **Reforms in the Budgetary Process:** Establishing a Budget Office in Parliament and preparing Pre-Budget Consultation Papers can help integrate gender perspectives into the budget process.

The Ministry of Finance plays a crucial role in making the budget process more transparent and inclusive, allowing representatives and citizens to participate actively in shaping budgets.

Q7: What are the main issues associated with the current Agri subsidy system in India? Suggest measures to reduce these challenges and make these subsidies more beneficial. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with introducing in brief about status of Agricultural subsidies in India.

Body: Address the demand by discussing the main Issues with the Current Agri Subsidy System and measures to address those issues/challenges.

Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Agricultural subsidies in India are vital for farmer support but have led to economic inefficiencies, ecological imbalances, and other negative consequences. India's farm input subsidies spike to \$48.13 billion in 2022-23

Main Issues with the Current Agri Subsidy System

- **Economic Inefficiency:** Subsidies often do not reach the intended beneficiaries, leading to resource wastage and reduced productivity. For example: Fertilizer subsidies primarily benefit large farmers rather than smallholders.
- **Resource Inefficiency:** Subsidies discourage efficient resource use. Example: Power subsidies causing water wastage in agriculture.
- **Corruption and Leakages:** Farm subsidies often suffer from corruption and misuse, failing to reach intended beneficiaries.
- **Neglect of Agricultural Research:** High subsidies divert funds from essential areas like agricultural research and farmer producer organizations (FPOs).
- **Environmental Harm:** Subsidies for chemical fertilizers promote excessive use, degrading soil health and polluting water sources.

- **Ecological Imbalance:** Current subsidies favor certain crops, depleting water resources. Example: Sugarcane in water-scarce regions like Maharashtra.
- **Overuse of Fertilizers:** Fertilizer subsidies lead to soil degradation. Example: Excessive nitrogen use reduces soil health.

Measures to Reduce Challenges

- **Promote Agricultural Infrastructure and Research:** Invest in agricultural research to increase efficiency and productivity. For example: Kelkar Committee recommendations.
- **MSP Reforms:** Align MSP with agro-climatic conditions. For example: Include millets like jowar, bajra, and ragi under MSP.
- **Digitalization of Subsidy Allocation:** Implement end-to-end digitalization to prevent leakages and corruption.
- **Decentralized Procurement:** Procure local crops to support regional agriculture. For example: Red rice in Uttarakhand, jowar in Rajasthan.
- **Targeted Subsidies:** Rationalize and target subsidies to benefit small farmers and promote resource efficiency. For example: Evidence-based delivery, farm credit.
- **Re-evaluate Populist Measures:** Assess and potentially phase out measures like loan waivers linked to vote bank politics.

To enhance the benefits of agricultural subsidies, India needs to rationalize the subsidy regime, promote research, and sustainable agricultural practice.

Q8: 'Entrepreneurial Spirit can transform agriculture sector into a profitable enterprising sector.' In this context, discuss how agriculture marketing can facilitate this transformation. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with introducing in brief about meaning of Entrepreneurial spirit in agriculture.

Body: Address the demand by discussing how agricultural marketing will transform the agriculture.

Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Entrepreneurial spirit can revolutionize the agriculture sector, making it profitable by leveraging digital technologies to enhance agriculture marketing, production, risk management, and sustainability.

Agriculture Marketing for Transformation:

- **Improved Market Access:** Agriculture marketing platforms connect farmers directly to consumers, reducing intermediaries and increasing farmers' profit margins.
 - Example: E-NAM (Electronic National Agriculture Market) facilitates transparent trading.
- **Enhanced Value Addition:** Processing and branding agricultural products increase their market value, attracting higher prices and broader markets.
 - Example: Amul's dairy products transformed milk into high-value goods.
- **Technological Integration:** Digital marketing tools help farmers reach a wider audience and optimize pricing through real-time data analysis.
 - Example: Mobile apps like Kisan Suvidha provide market information to farmers.
- **E-commerce Integration:** Integration of e-commerce in agriculture enables direct sales to consumers, enhancing profit margins and customer trust. For example: Platforms like BigBasket source produce directly from farmers.
- **Infrastructure Development:** Investment in storage, transportation, and cold chain logistics reduces post-harvest losses and ensures quality maintenance.
 - Example: Mega Food Parks scheme enhances infrastructure for processing and storage.
- **Access to Finance:** Entrepreneurial ventures in agriculture require funding for innovation and scaling operations, accessible through agricultural marketing platforms.
 - Example: NABARD's financial support for agribusiness projects.
- **Policy Support:** Government policies encouraging private investments and public-private partnerships in agriculture marketing foster an entrepreneurial environment.

➤ Example: Agricultural Produce Market Committee (APMC) reforms facilitate direct selling.

Agriculture marketing, by improving market access, value addition, technology use, infrastructure, finance, and policy support, empowers entrepreneurs to transform agriculture into a profitable enterprise.

Q9: Discuss the effects of irrigation systems on the depletion of groundwater resources in India. Highlight the role of modern irrigation technologies in promoting sustainable groundwater management.

(10 Marks, 150 Words)

Approach to the answer:

Introduction: Start with introducing in brief about irrigation system.

Body: Address the demand by highlighting effects of irrigation system on the depletion of ground water then mention role of modern irrigation technologies in promoting sustainable groundwater management.

Conclusion: Conclude with summarizing the topic.

Answer: According to the Central Ground Water Board (CGWB), India extracts approximately 230 billion cubic meters of groundwater annually for agricultural irrigation.

Effects of irrigation systems on the depletion of groundwater resources:

- **Over-extraction of Groundwater:** The irrigation sector is the largest consumer of groundwater in India, accounting for 89% of the total groundwater extracted, as per CGWB studies.
- **Declining Water Tables:** Overexploitation of groundwater resources has caused a decline in water tables, requiring deeper wells to access water. Approximately 30% of the monitored wells have shown a groundwater level drop, mostly between 0 and 2 meters (CGWB 2021 data).
- **Water Losses:** Inefficient irrigation methods, such as open canals and flood irrigation, lead to significant water losses due to evaporation, seepage, and runoff, further straining the groundwater supply.
- **Aquifer Depletion and Quality Degradation:** Unsustainable extraction and over-pumping can lead to aquifer depletion and degrade water quality. For instance, over-pumping in coastal regions can cause saltwater intrusion, contaminating groundwater.
- **Estimated Groundwater Depletion:** India's estimated groundwater depletion ranges between 122 and 199 billion cubic meters.

Role of Modern Irrigation Technologies in Promoting Sustainable Groundwater Management

- **Optimizing Water Use Efficiency:** Modern irrigation systems, such as drip irrigation, deliver water directly to plant roots, minimizing losses due to evaporation or runoff.
- **Conjunctive Use of Water Sources:** These technologies combine surface and groundwater use, balancing demand and conserving resources.
- **Soil Productivity:** Efficient irrigation conserves water, limits weed growth, and optimizes soil moisture, reducing groundwater demand.
- **Data-Driven Decisions:** Sensors and data analytics improve farm management, optimizing irrigation scheduling and resource use, thereby reducing groundwater reliance.

By adopting modern irrigation technologies, India can improve water use efficiency, reduce groundwater depletion, and work towards a sustainable and water-secure future. Lessons from other countries' successful implementations of these technologies can guide India's efforts to optimize water use in agriculture.

Q10. Describe the key pillars of Climate Smart Agriculture (CSA). How does the implementation of CSA contribute to achieving the Sustainable Development Goals (SDGs) 2030? (10 marks, 150 words)

Approach to the answer:

Introduction: Start with definition of Climate-smart agriculture (CSA).

Body: Address the demand by discussing Key Pillars of Climate Smart Agriculture (CSA) followed by Contribution of CSA to Achieving SDGs.

Conclusion: Conclude with summarizing the topic.

Answer: FAO defines Climate-smart agriculture (CSA) as “agriculture that sustainably increases productivity, enhances resilience (adaptation), reduces/removes GHGs (mitigation) where possible, and enhances achievement of national food security and development goals”

Key Pillars of Climate Smart Agriculture (CSA):

- **Sustainable Increase in Agricultural Productivity and Income:** Adopt high-yield, drought-resistant crops, use efficient irrigation (drip, sprinkler), and apply precision agriculture.
- **Adaptation and Resilience to Climate Change:** Develop climate-resilient crops, diversify farming systems, and improve soil health with conservation agriculture.
- **Reduction of Greenhouse Gas Emissions:** Use low-emission practices, improve livestock management, and enhance manure management with biogas production.

Contribution of CSA to Achieving SDGs:

- **Zero Hunger (SDG 2)** - Increases agricultural productivity, ensuring food security.
- **Good Health and Well-being (SDG 3)** - Promotes sustainable food systems for better nutrition.
- **Gender Equality (SDG 5)** - Empowers women with access to resources and technologies.
- **Clean Water and Sanitation (SDG 6)** - Promotes water-saving technologies and sustainable water management.
- **Sustainable Cities and Communities (SDG 11)** - Supports urban agriculture, enhancing city food security.
- **Climate Action (SDG 13)** - Mitigates climate change and builds resilience.
- **Life on Land (SDG 15)** - Promotes sustainable land management, protecting biodiversity.

Climate Smart Agriculture is a transformative approach that not only enhances agricultural productivity and resilience but also plays a crucial role in mitigating climate change. To achieve the ambitious goals of the 2030 Agenda, it is imperative to scale up CSA initiatives and invest in research, capacity building, and policy support.

Q11: ‘Bold economic reforms are challenging, yet essential.’ In this context, analyze the reasons for the success of the 1991 economic reforms in India and discuss the challenges in implementing bold economic reforms today.
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about 1991 economic reforms and link it with current challenges faced by economic reforms.

Body: Address the demand by highlighting key reasons for the success of 1991 economic reform followed by Challenges in Implementing Bold Economic Reforms Today.

Conclusion: Conclude with summarizing the topic or a way forward.

Answer: The 1991 economic reforms known as liberalization, privatization, and globalization (LPG) reforms transformed Indian economy, leading to unprecedented growth. Today, bold economic reforms remain crucial but face significant challenges.

Reasons for the Success of the 1991 Economic Reforms:

- **Economic Crisis:** Severe balance of payments crisis necessitated immediate action.
 - For example: India's foreign exchange reserves were depleted to just two weeks' worth of imports.
- **Political Will:** Strong political leadership and consensus.
 - For example: Prime Minister P.V. Narasimha Rao and Finance Minister Dr. Manmohan Singh's decisive roles.
- **International Support:** Assistance from the International Monetary Fund (IMF).
 - For example: IMF's structural adjustment program provided necessary financial support.

- **Globalization Trends:** Alignment with global economic trends.
 - For example: Liberalization of trade policies facilitated integration with the global economy.
- **Domino Effect:** Deregulation dismantled the 'License and Permit Raj,' spurring economic growth.

Challenges in Implementing Bold Economic Reforms Today

- **Political Resistance:** Fragmented political landscape hinders consensus.
 - For example: Opposition to agricultural reforms led to nationwide protests in 2020-21; Union-state disagreements hinder reforms, e.g., electricity and land reforms.
- **Social Backlash:** Reforms may face public opposition due to perceived adverse impacts.
 - For example: Labor law reforms faced resistance from trade unions.
- **Economic Disparities:** Increasing inequality poses challenges to implementing reforms.
 - Example: High-income disparity may lead to social unrest.
- **Bureaucratic Hurdles:** Administrative inertia and corruption can delay reforms.
 - For example: Slow implementation of GST due to bureaucratic complexities.
- **Global Uncertainties:** Economic policies influenced by global economic instability.
 - For example: Non-functional WTO.
- **Environmental Concerns:** Balancing growth with sustainability, e.g., Forest (Conservation) Amendment Act 2023.

Measures to Overcome Challenges:

- **Political Consensus:** Avoid 'competitive populism' and build consensus.
- **Enabling Ecosystem:** Invest in skilling and infrastructure for the 4th Industrial Revolution, e.g., PM-Gati shakti Masterplan.
- **Inclusive Growth:** Design reforms to support MSMEs.
- **Reformed Multilateralism:** Reform global institutions like the UN and WTO.
- **Environmental Sustainability:** Ensure green policies and effective implementation, e.g., green growth in Budget 2023.

While bold economic reforms are essential for sustained growth, addressing political, social, and economic challenges is crucial for successful implementation in today's context to achieve India's goal of becoming a developed nation by 2047.

Q12: Despite several efforts aimed at improving the fiscal condition of states in India, there are still numerous challenges that need to be addressed. Elaborate (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Fiscal Health of States in India.

Body: Address the demand by highlighting recent steps to improve states fiscal health followed by key issues affecting states fiscal health then mention recommendations to improve it.

Conclusion: Conclude with summarizing the topic.

Answer: The fiscal health of Indian states has been a longstanding concern, exacerbated during the pandemic years. In 2020-21, the gross fiscal deficit (GFD) of states reached 4.1% of GDP, and the debt-to-GDP ratio rose to 31.1%, significantly exceeding the 20% recommended by the Fiscal Responsibility and Budget Management (FRBM) Review Committee.

Recent Initiatives to Improve States' Fiscal Health

- **Special Assistance Scheme:** This Union government provides states with a ₹1,00,000 crore interest-free loan for 50 years to support capital investments.
- **Reform-Linked Additional Borrowing:** States can access an additional borrowing space of 0.5% of their Gross State Domestic Product (GSDP) if they implement power sector reforms.
- **Inclusion of Off-Budget Borrowings:** State debt positions now include off-budget borrowings, such as those taken by state government entities and special purpose vehicles, ensuring a more accurate representation of fiscal health.

- **Enhanced Revenue Collection:** Improvement in revenue collections, particularly through the implementation of the State Goods and Services Tax (SGST) and excise duties, has led to increased tax devolution from the central government.

These measures have led to a budgeted reduction in states' GFD to 3.4% of GDP and a decrease in the debt-to-GDP ratio to 29.5% for 2022-23. While these improvements are notable, the fiscal indicators still fall short of the FRBM Review Committee's recommendations.

Key Issues Affecting States' Fiscal Health

- **Increasing Share of Cess and Surcharge:** The share of cess and surcharges in the Centre's gross tax revenue has risen from around 8% in 2011-12 to 28% in 2021-22, thereby reducing the funds available for tax devolution to the states and impacting their revenues.
- **Dominance of Committed Expenditures:** In 2022-23, states spent 54% of their revenue receipts on salaries, pensions, and interest payments, limiting funds for developmental projects.
- **Lower States' Own Revenue:** States have low own revenues due to losses in State PSUs, weak discom finances, inadequate user charges, and challenges like high administrative costs, lack of innovation, and economic shocks.
- **High Revenue Expenditure:** Revenue expenditure comprises 80-90% of total expenditure in most states, limiting the funds available for capital investments.
- **Growing Demand for Public Services:** Rising demands for health, education, and infrastructure investment further strain state finances, especially in the aftermath of the Covid-19 pandemic.
- **Other Contributing Factors:** Populist measures, such as farm loan waivers and the reintroduction of old pension schemes by various state governments, exacerbate fiscal stress.

Recommendations for Improvement

- **Debt Consolidation:** State governments should prioritize debt consolidation to achieve sustainable fiscal health.
- **Creating a Favorable Environment for Private Investment:** States should focus on fostering a conducive environment for private sector growth, which is vital for attracting investment.
- **Enhancing Inter-State Trade and Commerce:** States should encourage higher levels of inter-state trade and commerce to leverage the benefits of spillover effects from state capital expenditures across the country.
- **Diversified Revenue Generation:** States should explore innovative financing mechanisms and diversify revenue sources, such as non-tax avenues, to reduce reliance on central transfers.

Addressing the ongoing fiscal challenges faced by Indian states requires continuous reform efforts, innovative solutions, and strong governance to ensure sustained economic development and equitable resource distribution.

Q13: What are the objectives of the Production Linked Incentive (PLI) schemes in India? Identify the primary challenges in its implementation and suggest measures to overcome them. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Production Linked Incentive (PLI) schemes.

Body: Address the demand by highlighting objectives of PLI schemes followed by challenges faced by PLI schemes.

Conclusion: Conclude with summarizing the topic or a way forward.

Answer: The Production Linked Incentive (PLI) schemes aim to boost domestic manufacturing, attract investment, enhance export capabilities, and reduce import dependency in various sectors. These schemes incentivize companies to increase production by providing financial incentives based on incremental sales.

The PLI scheme has the following objectives:

- **Domestic Manufacturing:** By offering incentives based on incremental turnovers, the scheme encourages the expansion of manufacturing capacities.
- **Import Substitution:** Domestic production can reduce import dependency in sectors like solar panels and pharmaceutical APIs.
- **Employment Generation:** The growth of the manufacturing sector is expected to generate employment opportunities, with an estimated 60 lakh new jobs between 2021-22 and 2026-27.
- **Attracting Foreign Investment:** This attractive scheme increased foreign investment in manufacturing. For example, a 76% increase in FDI in the manufacturing sector in FY2022 over FY2021.
- **Innovation and Technology Transfer:** Establishing a manufacturing base in India by MNCs promotes innovation and technology transfer.
 - For example, include Foxconn as an anchor investor in the semiconductor sector.

The PLI scheme faces several challenges:

- **High Initial Investment:** Manufacturers need substantial investments to qualify for incentives, which can be a barrier for smaller domestic units and MSMEs.
- **Lack of Skilled Workforce:** The success of the PLI scheme depends on a skilled workforce, but only 46% of graduates in India are employable, according to the India Skills Report.
- **Supply Chain Disruptions:** Russia-Ukraine conflict have impacted global manufacturing, and India remains import-dependent for many intermediate goods and raw materials.
- **Structural Issues:** Infrastructure deficits and slow clearances deter investments in manufacturing.
 - For example, Vietnam has become a preferred investment destination due to low labor costs and FTAs (like RCEP).
- **Technology Transfer Issues:** There are fears that India may become an assembly hub rather than a manufacturing hub.
 - For example, cancellation of the Foxconn-Vedanta deals due to a lack of technology transfer.

Way Forward:

- **Workforce Development:** Skilling, reskilling, and upskilling the workforce to meet changing job demands.
- **Strengthening IPR Protection:** Promoting innovation and boosting investor confidence by enhancing Intellectual Property Rights (IPR) protection.
- **Leveraging Traditional Advantages:** Utilizing India's traditional strengths, such as technical textiles, a farm-to-fork approach in the food-processing industry, etc.
- **Addressing Infrastructure Deficit:** Frontloading infrastructure spending, for instance, through plug-and-play infrastructure like Dholera SEZ.

A combination of targeted policy interventions and proactive engagement with relevant stakeholders is essential to realize the full potential of the PLI scheme, thereby strengthening India's march towards achieving the goal of Atmanirbhar Bharat.

Q14: Discuss the challenges hindering India's export competitiveness and suggest measures to enhance the role of exports in the country's economic growth. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Status of Indian exports.

Body: Address the demand by highlighting factors inhibiting India's export competitiveness followed by measures to enhance India's export growth.

Conclusion: Conclude with summarizing the topic.

Answer: Despite contributing around 3.1% to the world's GDP, India's share of global exports is only 1.6%. While the recent surge in exports to \$418 billion in FY 2021-22 is commendable, a significant gap exists between India's export potential and actual performance across many sectors.

Strong exports have been key drivers of development for countries like Japan, South Korea, China, and Thailand.

Factors Inhibiting India's Export Competitiveness:

- **High Tariffs on Intermediate Goods:** Nearly 70% of anti-dumping duties in India are imposed on intermediate goods, which are essential inputs for finished export products. This raises production costs thereby reducing export competitiveness.
- **Traditional Export Basket:** India's exports largely consist of non-cutting-edge products, with 70% targeting 30% of world trade in items with declining global demand.
- **Restrictive Business Environment:** Complex trade regulations, red tape, intricate taxation laws, restrictive labor laws, and high non-tariff barriers such as administrative fees and labeling requirements hinder business operations and exports.
- **Limited Access to Finance:** Many Indian exporters, especially SMEs, face challenges in accessing affordable credit and trade finance. This constrained financial support limits their ability to invest in technology and improve infrastructure, thus curtailing their export potential.
- **High Logistics Costs:** The logistics cost in India stands at approximately 16% of GDP, significantly higher compared to about 10% in China and 8% in the US and Europe.
- **One-Size-Fits-All Approach:** India's export strategy lacks sector-specific and market-specific plans, limiting its ability to fully capitalize on potential opportunities in different sectors.
- **Competition and Market Dynamics:** Tariff wars and geopolitical tensions impacts buyer preferences and demand patterns, complicating foreign trade strategies.

Measures to Enhance India's Export Growth:

- **Simplifying Regulatory Frameworks:** Implementing digital platforms for compliance and documentation which can expedite procedures, enabling faster market access.
- **Increasing Access to Finance:** Expanding avenues for affordable credit and incentivizing private banks to lend to exporters can strengthen financial resilience.
- **Decentralized Export Strategy:** India should adopt a more decentralized and state-specific export policy. Utilizing the NITI Aayog's Export Preparedness Index (EPI) can foster competition among states and encourage the adoption of export promotion policies tailored to local strengths.
- **Aggressive Participation in Global Value Chains (GVCs):** India should increase its participation in GVCs and leverage its working-age population in labor-intensive manufacturing, especially in the assembly of network products.
- **Enhancing the Export Ecosystem:** Efficient implementation of trade facilitation initiatives such as Remission of Duties or Taxes on Export Products (RoDTEP), PM Gati Shakti, Trade Infrastructure for Export Scheme (TIES), the new Labor Codes, and Market Access Initiative by the Export Promotion Councils is crucial.
- **Strengthening Domestic Capacity:** Promoting domestic manufacturing through measures like lowering corporate tax rates and introducing Production Linked Incentive (PLI) schemes in key sectors can enhance India's export base.
- **Moving Away from Protectionism:** India should embrace preferential trade agreements and regional trading arrangements. The country needs to reorient its trade policy to benefit from the China-plus-one strategy, which seeks to diversify supply chains away from China.

These measures will help India achieve ambitious targets of \$1 trillion in merchandise exports by 2027-28 and \$1 trillion in services exports by 2030, thereby advancing towards the goal of becoming a \$5 trillion economy.

Q15: "Land reforms are critical to achieving farm prosperity in India." In this context, critically analyse the challenges associated with implementing pending land reforms in India. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Status of land holding in India.

Body: Address the demand by highlighting the challenges impede the effective implementation of pending land reforms followed by way forward to address those issues.

Conclusion: Conclude with summarizing the topic.

Answer: India boasts the second-largest arable land area globally, with around 159.7 million hectares. Currently, landless, marginal, and small farmers make up 93.7% of India's agricultural workforce. The Socio-Economic and Caste Census 2011 further highlights that 56% of rural households do not own any agricultural land, despite heavy reliance on agriculture.

Challenges impede the effective implementation of pending land reforms:

- **Historical Challenges:** The legacy of colonial land led to extreme disparities, where a small percentage of landowners controlled a majority of agricultural land, while vast population remained landless or owned minimal land.
- **Bureaucratic Inefficiencies:** The presence of corruption and collusion between bureaucrats and landlords further exacerbates these issues, often leading to prolonged legal disputes and inadequate enforcement of reforms.
- **Resistance from Vested Interests:** Large landowners and political groups that benefit from the status quo often resist changes to land ownership policies. Such opposition slows down equitable distribution of land to support farmer prosperity.
- **Legal Complexities:** Issues such as inadequate land records, ambiguous definitions of land use, and varying laws across states make it challenging to implement land reform.
- **Lack of Political Will:** Weak political commitment led to incomplete or ineffective reforms, leaving land inequalities unaddressed and undermining farm prosperity objectives.

Way Forward:

- **State Control over Land:** As land is a 'state subject' under the 7th Schedule of the Constitution. There is a debate about whether land reforms should become a 'central subject'.
- **Land-Leasing and Contract Farming:** Introducing innovative land-leasing and contract farming methods can help consolidate land holdings.
- **Redistribution and Ceiling Limits:** There's a need to redistribute surplus land, revise land ceiling limits in irrigated areas, and accurately record and distribute land.
- **Land Titling and Modernization:** Digitizing land records and modernizing land titles can prevent further fragmentation of land. Awareness campaigns for farmers should accompany this.
- **Progressive Land Tax:** Implementing a progressive tax based on land ownership size could encourage the sale of larger landholdings.
- **Empowering the National Council for Land Reforms:** Giving mandatory powers to the National Council for Land Reforms would strengthen the implementation of land policies and address related issues effectively.

Only through a comprehensive and inclusive approach can land reforms truly contribute to farm prosperity in India.

Q16: 'The Green Revolution created a situation of water stress in farming.' Examine the statement and suggest measures to make agriculture sustainable and future ready. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about green revolution.

Body: Address the demand by highlighting how the green revolution created a situation of water stress followed by implication of water stress.

Conclusion: Conclude with summarizing the topic or with a way forward.

Answer: The Green Revolution, while significantly boosting agricultural productivity, inadvertently created a situation of water stress in farming by treating water as an unlimited resource and promoting water-intensive crops.

Green Revolution created a situation of Water Stress:

- **High-Yield Varieties (HYVs) Demand More Water:** Green Revolution introduced HYVs seeds of rice and wheat which requiring extensive irrigation, stressing groundwater reserves.
- **Water as an Unlimited Resource:** The Green Revolution focused on increasing land productivity, assuming water availability was infinite. This led to inefficient water usage.
 - For example: Indian farmers use 2-4 times more water than Chinese farmers for the same crops.
- **Poor Water Use Efficiency:** Free water and electricity for irrigation such as Subsidized electricity for irrigation resulted in over-extraction of groundwater.
- **Promotion of Water-Intensive Crops:** Price support encouraged the cultivation of water-intensive crops beyond suitable agro-climatic zones.
 - For example: Drought-prone areas like Vidarbha and Rajasthan grow sugarcane, exacerbating water stress.
- **Export of Water-Intensive Crops:** Punjab and Haryana face groundwater depletion due to exporting water-intensive crops like rice.

Implications of Water Stress

- **Income Loss for Farmers:** VoxDev reported a 25% income reduction for farmers with dried wells.
- **Climate Vulnerability:** Economic Survey 2020 noted a 60% higher vulnerability in unirrigated lands due to climate change.
- **GDP Loss:** NITI Aayog predicts a 6% GDP loss by 2050 due to water scarcity.
- **Food Security threat:** All four pillars of food security—availability, access, utilization, and stability—are threatened.
- **Social and Developmental Impact:** Water stress can negatively impact development indicators such as Increased infant mortality rates, disease burden, etc.

Measures for Sustainable Agriculture

- **Reduce Water Wastage:** Rationalize electricity subsidies, adopt micro-irrigation, and use auto-cutoff sensors.
- **Regulate Water Usage:** A Central Groundwater Authority to manage water resources effectively.
- **Crop Suitability:** Promote crops based on agro-climatic zones such as encourage millet cultivation in arid regions.
- **Traditional and Modern Conservation Methods:** Combine traditional water conservation techniques with modern methods.
 - For example: Use johads for water storage alongside fog catchers and greywater reuse.
- **ICT and Precision Agriculture:** Utilize technology to optimize water use.
 - For example: Implement precision agriculture and greenhouse technologies.

Enhancing water productivity through a shift towards "more crop per drop" agriculture is vital for ensuring food security and achieving Sustainable Development Goals.

Q17: Highlight the role of the Food Corporation of India (FCI) in ensuring food security through the Public Distribution System (PDS). How do the issues within the FCI affect the efficiency and effectiveness of the PDS? (15 marks, 250 words)

Approach to the answer:

Introduction: Start with introducing in brief about FCI and PDS in India.

Body: Address the demand by highlighting Role of FCI in Ensuring Food Security through PDS followed by issues within FCI affecting PDS efficiency and effectiveness then suggest way forward.

Conclusion: Conclude with summarizing the topic.

Answer: The Food Corporation of India (FCI) plays a pivotal role in ensuring food security through the Public Distribution System (PDS) by handling procurement, storage, and distribution of food grains.

Role of FCI in Ensuring Food Security through PDS

- **Procurement:** FCI procures rice and wheat from farmers at minimum support prices (MSP) to ensure stable income and price stability.
- **Storage:** FCI stores procured grains in its warehouses, maintaining a buffer stock to address food shortages or during emergencies.
 - For example: During COVID-19, buffer stocks were used to provide free rations.
- **Distribution:** FCI allocates food grains to states for distribution through PDS, ensuring subsidized food reaches the poor.
 - For example, states like Tamil Nadu known for its Targeted PDS (TPDS).
- **Maintaining Buffer Stocks:** FCI maintains buffer stocks to ensure food security during emergencies.

However, issues within FCI affecting PDS efficiency and effectiveness:

- **Rice-Wheat Bias:** FCI's focus on rice and wheat leads to neglect of nutritious crops like millets, impacting nutrition security.
- **Storage Infrastructure:** Poor storage facilities result in spoilage and wastage of grains.
 - Inadequate storage led to an estimated around ₹92,000 crores value of food wastage every year.
- **Operational Inefficiencies:** Delays in procurement and distribution processes lead to inefficiencies, affecting timely delivery and quality of food grains in PDS.
- **Corruption and Leakages:** Corruption and leakages within FCI and PDS channels lead to diversion of food grains such as unauthorized sale of PDS grains in open markets
- **State Role:** Limited implementation of decentralized procurement by states hampers efficient distribution.
 - Only 15 states have adopted decentralized procurement for rice.

Measures to address FCI and PDS Challenges

- **Diversification:** Shift FCI's focus to include millets and vegetables in procurement and distribution.
- **Decentralized Procurement Scheme:** Gradually transfer procurement responsibilities for rice and wheat to states, as recommended by the Shanta Kumar Committee.
- **Universalization and Assured Procurement:** Broaden PDS coverage and ensure procurement for all crops, while narrowing the MSP crop-basket to manage fiscal sustainability.
- **Private Sector Involvement:** Enhance storage infrastructure through private sector participation, like the Private Entrepreneurs Guarantee (PEG) scheme.
- **Technological Interventions:** Use technology to reduce leakages, improve price realization, and address inclusion-exclusion errors.
 - For example: GPS tracking and direct-benefit transfers like e-RUPI.

Reforms in both FCI and PDS are crucial. Enhancing capacity, decentralizing functions, and leveraging technology will address inherent challenges and ensure effective food security.

Q18: Highlight the principles of regenerative agriculture work. How does it address the challenges faced by agriculture?
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about regenerative agriculture.

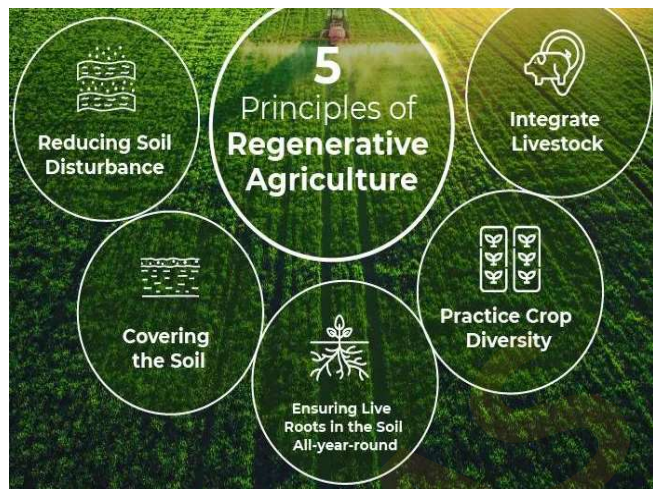
Body: Address the demand by highlighting principles of regenerative agriculture followed by key issues facing the agricultural sector.

Conclusion: Conclude with summarizing the topic or with a way forward.

Answer: Regenerative agriculture is a sustainable farming approach that emphasizes conservation and rehabilitation of the ecosystem.

Principles of regenerative agriculture:

- **Minimize Soil Disturbance:** Practices like avoiding ploughing and minimizing the use of synthetic fertilizers to avoid damages to the micro-flora and fauna integral to soil health.
- **Keep the Soil Covered:** Maintaining soil cover with living plants or crop residues protects soil from damage caused by rainfall and extreme temperatures. This also enhances water absorption and retention, benefiting soil moisture levels.
- **Maintain Living Roots in the Soil:** Having living plant roots in the soil year-round is crucial for supporting the underground ecosystem.
- **Maximize Plant Diversity:** Diverse plant populations enhance soil resilience, improve nutrient cycling, and reduce pest and disease pressures.
- **Integrate Livestock:** Grazing animals can enhance soil fertility and contribute to carbon sequestration, playing a vital role in nutrient cycling within the ecosystem.



Key issues facing the agriculture sector:

- **Mono-cropping and Soil Degradation:** The focus on the rice-wheat system has led to mono-cropping, disrupting natural cropping patterns. This practice has caused soil degradation, stubble burning, etc.
- **Overuse of Fertilizers:** Fertilizer subsidies have led to excessive use of urea, causing soil and water pollution through leaching into groundwater.
- **Groundwater Depletion:** The MSP system and free electricity have incentivized water-intensive paddy cultivation in water-scarce regions like Punjab and Haryana, leading to groundwater depletion.
- **Nutrient Deficiencies:** The neglect of nutritious pulses and millets in diets has led to rising nutrient deficiencies.
- **Loss of Food and Cultural Diversity:** Changing food habits have led to the declining popularity of local cuisines, impacting both food and cultural diversity.

Regenerative Agriculture addresses these issues:

- **Community Integration:** Regenerative agriculture integrates the community into agricultural supply chains, revitalizing the rural economy. For example, promoting Farmer Producer Organizations (FPOs).
- **Climate Adaptation and Mitigation:** Practices like agroforestry help in carbon sequestration and improving soil organic carbon, aiding in climate change adaptation and mitigation.
- **Soil Quality and Cost Reduction:** Alternative practices like Zero-Budget Natural Farming (ZBNF) and growing drought-resistant crops like millets can improve soil quality and reduce input costs.
- **Integrated Farming:** Combining different farming practices, such as paddy-fish cultivation and poultry farming, enhances nutrient recycling and increases farm income. Techniques like drip irrigation and precision agriculture can improve farm yield.
- **Food Security:** Regenerative agriculture ensures intergenerational equity in farming systems, preventing future food crises by protecting soil health and improving the nutrient content of food crops.
- **Economic Viability for Small Farmers:** For small and marginal farmers, regenerative practices like organic farming can significantly reduce input costs, making them economically viable.

Transitioning to regenerative agriculture is crucial for achieving a balance between environmental sustainability, food security, and community well-being.

Q19: How does livestock rearing impact India's agriculture and economy? Describe major government initiatives supporting this industry and assess the successes and shortcomings of livestock rearing in the country.
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Start with introducing in brief about Livestock rearing.

Body: Address the demand by highlighting how it plays a significant role in India's agricultural and economic landscape followed by Key Government Initiatives for Livestock Sector Growth then mention the successes and Challenges in the Livestock Sector.

Conclusion: Conclude with summarizing the topic.

Answer: Livestock rearing is the practice of breeding and raising domestic animals primarily for economic purposes. It is a vital component of agriculture, contributing to food production, income generation, and the overall economic stability of rural communities.

It plays a significant role in India's agricultural and economic landscape:

- **Income Security:** Livestock provides a stable source of income, particularly for small and marginal farmers, and offers a cushion against crop failures and price crashes.
- **Nutritional Security:** It enhances the availability and access to nutrition by supplying affordable products like milk, curd, meat, and eggs.
- **Socio-Economic Empowerment:** The sector empowers marginalized communities, with women constituting over one-third of the workforce in livestock rearing.
- **Sustainable Agriculture:** Livestock rearing contributes to sustainable agriculture as it is less affected by climate change than food crops. Livestock waste is utilized in practices like organic farming and Zero Budget Natural Farming (ZBNF).
- **Food Processing Industry:** It supports downstream value addition through traditional food processing methods, such as making curd and butter.
- **Textile and Leather Industry:** It serves as an upstream sector for the employment-intensive textile and leather industries, contributing to products like woolen knitwear.

Key Government Initiatives for Livestock Sector Growth:

- **National Livestock Mission (NLM):** Aims to enhance employment generation through entrepreneurship and improve livestock breeds.
- **Animal Husbandry Infrastructure Development Fund (AHIDF):** Encourages investment from MSMEs, FPOs, and entrepreneurs.
- **National Animal Disease Control Program (NADCP):** Protects livestock from diseases like Foot and Mouth Disease (FMD) and brucellosis through vaccination.
- **National Artificial Insemination Program (NAIP):** Enhances milk production using low-cost breeding technologies to improve genetic merit.
- **Rashtriya Gokul Mission (RGM):** Promotes bovine breeding to conserve and develop indigenous breeds.
- **Pashu Aadhar Yojna (PAY):** Tracks livestock for disease control and productivity monitoring.

Successes and Challenges in the Livestock Sector:

Successes:

- **Global Production:** India is the largest producer of milk globally (23% of global milk production), ranks 3rd in egg production, and 8th in meat production.
- **Sector Growth:** The livestock sector has a high growth rate of 8%, with its share in agriculture GVA increasing from 24% to 31% between 2018 and 2022.
- **Income Contribution:** The sector contributes 16% of the income for small farmers.
- **Dairy Sector Impact:** Over 8 crore farmers rely directly on the dairy sector for their livelihoods.
- **Livestock Census:** Regular livestock censuses track the population, including 85 crore poultry, 30 crore bovines, 15 crore goats, and 7.4 crore sheep, as per the 20th Livestock Census.

Challenges:

- **Low Productivity:** Inadequate fodder quality results in low productivity, with Indian cattle producing 1777 kg of milk per year compared to the world average of 2699 kg.
- **Disease Prevalence:** Diseases like Lumpy Skin Disease (LSD), black quarter, and influenza continue to affect cattle.
- **Unscientific Handling and Resistance:** Issues such as unscientific handling and rising antimicrobial resistance hinder export potential.
- **Skilled Workforce Shortage:** There is a shortage of skilled human resources, including registered veterinarians.
- **Accessibility and Affordability:** Despite availability, livestock products remain unaffordable for many poor households.
- **Protectionism:** The sector seeks protection from international competition, as evidenced by India's refusal to sign the RCEP agreement due to concerns about the domestic livestock industry.

Addressing these issues through improved policies, infrastructure, and training is crucial for the sustainable growth and development of the livestock sector in India.

Q20: Discuss the factors contributing to the underperformance of the Food Processing Sector in India. Suggest measures to overcome these challenges. (15 marks, 250 words)

Approach to the answer:

Introduction: Start with introducing in brief about Food Processing Industry (FPI).

Body: Address the demand by highlighting factors contributing to underperformance of FPI in India followed by measures to overcome challenges in Food Processing Industry (FPI).

Conclusion: Conclude with summarizing the topic or a way forward.

Answer: The Food Processing Industry (FPI) involves the transformation of raw agricultural products into consumable food items through various processes such as cleaning, grading, sorting, packing, and value addition. Despite its potential and economic contributions, faces several challenges that hinder its growth and efficiency.

Factors Contributing to Underperformance of FPI in India

- **Perishability and Weather Dependency:** Perishability of vegetables and unpredictable weather conditions disrupt consistent quality and supply of raw materials.
 - For Example: Price volatility of tomatoes and potatoes.
- **Infrastructure and Technology Gaps:** Poor infrastructure such as inadequate roads and cold chain networks, disrupts supply chains and causes significant post-harvest losses.
 - For example: NITI Aayog reports annual post-harvest losses of Rs 90,000 crore.
- **Unorganized Segment Concentration:** A large proportion of FPI firms are unorganized, leading to inefficiencies and lower productivity in the sector.
 - For example: 75% of firms across all product categories are unorganized.
- **Ease of Doing Business Issues:** Bureaucratic hurdles and complex licensing policies hinder the establishment and operation of food processing industries.
 - For example: Numerous forms required for FSSAI licenses complicate business operations.
- **Dependence on APMCs:** Reliance on Agricultural Produce Market Committees (APMCs) increases costs and disrupts seamless linkage between farms and FPIs.
 - For Example: Additional costs are incurred due to APMC intermediaries.
- **Cultural Preferences:** Preference for homemade and fresh food over processed and canned food limits the market for processed food products.

Measures to Overcome Challenges in FPI

- **Risk Sharing Mechanisms:** Encourage private sector participation and fiscal incentives to create infrastructure for logistics, storage, and processing.

- Example: Pradhan Mantri Kisan Sampada Yojana (PMKSY) supports infrastructure development.
- **Skill Development:** Encourage academic and industry collaboration to offer courses in food packing, processing, and biotechnology for skilled manpower.
- **Regulatory Framework Development:** Implement regulatory frameworks for contract and corporate farming to ensure consistent and cost-effective raw material supply.
 - For example: Model Land Leasing Act 2016 and Model Contract Farming Act 2018.
- **Village-Level Procurement Centers:** Increase procurement centers at village level for perishables like fruits, vegetables, and dairy to ensure timely collection and processing.
 - For example: Upgrading 22,000 rural Haats into Gramin Agriculture Markets (GrAMS).
- **Conducive Policy Environment:** Government initiatives like extending Operation Greens Scheme and setting up grievance cells support industry resilience and growth.

Addressing the challenges in the FPI sector with strategic measures can unlock its potential, ensuring food security and economic growth of this sector.